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## RESEARCH ARTICLE

# Regime-Conditioned Horizon-Aware Mixture-of-Experts for Spatio-Temporal Wind Forecasting

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**ABSTRACT** Accurate short-term wind speed prediction (WSP) reduces operational risk in wind-integrated grids because turbine power amplifies wind-speed errors through a cubic dependence on velocity. Improved WSP therefore supports tighter reserve scheduling and more reliable dispatch under regime-dependent variability. Existing deep learning forecasters often rely on a single backbone or step-invariant fusion, which enforces uniform computation across forecast steps and can reduce robustness when dominant dynamics shift across regimes. Mixture-of-Experts (MoE) offers capacity allocation, yet many recent hybrids keep inference-time fusion fixed or use MoE mainly for offline optimization rather than step-wise, regime-conditioned routing. This work proposes Regime-Conditioned Horizon-aware Mixture-of-Experts (RCHA-MOE) for spatio-temporal forecasting. RCHA-MOE fuses three complementary experts that encode persistence, dilated multi-scale temporal structure, and locality-preserving recurrent dynamics, and it learns a Horizon-Conditioned Regime Gate (HCRG) that predicts step-wise mixture weights from the input window to enable regime-adaptive fusion at inference. Experiments on three hourly NASA POWER datasets under a fixed  $3 \times 3$  mesoscale sampling design show that RCHA-MOE achieves the best mean accuracy within this controlled benchmark, improving RMSE by 6.44%–6.93% and MAE by 7.49%–7.65% over the strongest baseline while increasing  $R^2$  by 0.77%–1.02%. Moreover, after power-curve mapping, RCHA-MOE reduces power and energy proxy errors, including in the ramp region where the power curve is most sensitive to wind-speed deviations.

**INDEX TERMS** Wind speed prediction, mixture-of-experts, adaptive gating, spatio-temporal learning, multi-step forecasting, renewable energy forecasting.

## I. INTRODUCTION

Wind energy increasingly supports low-carbon generation, yet regime-dependent variability complicates scheduling and dispatch because turbine power responds nonlinearly to wind-speed errors [1]. For a horizontal-axis turbine, the aerodynamic power relation is

$$P = \frac{1}{2} \rho A v^3 C_p, \quad (1)$$

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where  $P$  is power,  $\rho$  is air density,  $A$  is rotor swept area,  $C_p$  is the power coefficient, and  $v$  is wind speed [2]. Because the  $v^3$  term amplifies small forecast deviations into large power swings, accurate short-term wind speed prediction (WSP) remains a core requirement for operational reliability.

WSP methods typically fall into five classes that include physical models (PMs), statistical models (SMs), machine learning (ML), deep learning (DL), and hybrid models (HMs). PMs forecast wind by simulating atmospheric dynamics from first principles, most commonly through Numerical Weather Prediction (NWP) systems that solve

the primitive equations on a mesoscale grid with data assimilation and parameterizations for turbulence, radiation, and surface fluxes [3]. In wind-energy practice, the Weather Research and Forecasting (WRF) model often serves as an NWP backbone to produce gridded wind fields that support operational pipelines [4]. Because mesoscale resolution smooths terrain-induced speed-up and wake-relevant gradients, PM pipelines frequently add a microscale step by nesting higher-resolution domains or by downscaling WRF outputs into Computational Fluid Dynamics (CFD) solvers that represent local orography and roughness effects more explicitly [5]. In computational wind engineering, Reynolds-Averaged Navier–Stokes (RANS) and related closures estimate mean flow over complex terrain and built environments, which yields physically interpretable local adjustments when geometry dominates [6]. Moreover, response libraries and cost-efficient downscaling strategies inject microscale structure into NWP winds without running full transient CFD at every update [7]. However, PM accuracy typically demands fine grids, small time steps, multi-physics parameterizations, and frequent re-initialization as new observations arrive, which increases latency and hardware demand in short-term WSP.

SMs also learn from data, but they impose structured assumptions that keep calibration and updates inexpensive. Autoregressive (AR) baselines capture persistence [8], Autoregressive Moving-Average (ARMA) absorbs serially correlated shocks and measurement noise through a moving-average error term [9], and Autoregressive Integrated Moving-Average (ARIMA) introduces differencing when nonstationarity appears so the residual process better matches stationarity assumptions [10]. Seasonal ARIMA (SARIMA) adds seasonal operators to represent recurring cycles [11], while ARIMAX and SARIMAX incorporate exogenous drivers to reduce omitted-variable bias and improve short-term adaptation [12], [13]. These models often deliver strong short-term accuracy under operational constraints because they correct location-specific bias quickly and run cheaply. However, they usually rely on linear dynamics and stable innovation statistics, so wind ramps, regime shifts, and terrain-driven intermittency can violate assumptions and degrade performance unless the pipeline refits aggressively or adds richer structure.

Learning-based methods relax these constraints by learning nonlinear mappings from historical measurements and multivariate covariates, which reduces reliance on hand-crafted corrections and strengthens short-term accuracy in many datasets [14], [15]. Classical ML forms a compact ladder. Kernel regressors such as Support Vector Regression (SVR) and Kernel Ridge Regression (KRR) exploit nonlinear feature spaces and often outperform linear baselines when chaotic dynamics and multicollinearity matter [16], [17]. Ensembles and latent-factor regressions improve robustness by pooling weak learners and compressing correlated signals, yet they still depend on manual lag design and ad hoc spatial

inputs [18], [19]. Gradient-boosted trees then mature into strong tabular baselines for short-term WSP, but their accuracy remains sensitive to feature engineering and hyperparameter search under regime change [20], [21]. DL shifts the bottleneck from manual features to learned representations. Recurrent models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) capture longer temporal dependence [22], [23], convolutional designs such as Temporal Convolutional Networks (TCNs) scale receptive fields with stable gradients [24], and attention-based Transformers model global context through tokenization choices that improve multi-step learning when spatio-temporal coupling strengthens [25], [26], [27]. However, recent studies show that carefully designed hybrid models often outperform Transformer-based architectures in short-term predictions, especially when local spatio-temporal structure and regime effects dominate [1], [28].

HMs aim to improve robustness by combining complementary inductive biases rather than committing to a single backbone. Many HM pipelines decompose the signal before learning. Empirical Mode Decomposition (EMD) splits a nonstationary series into intrinsic mode functions that isolate oscillations at different scales [29], while Variational Mode Decomposition (VMD) extracts band-limited modes through constrained optimization and often reduces mode mixing [30]. Other HMs fuse architectures directly, for example by coupling convolutional feature extraction with recurrent memory in CNN BiLSTM models that capture local patterns and temporal persistence in one pipeline [31]. In parallel, physics-informed hybrids regularize predictors with physical structure, but simplified operators and weak priors can underconstrain solutions or introduce bias when the surrogate mismatches the data regime [1], [32]. Physics-informed ensemble learning has also been extended to city-center grid-cell temperature prediction during thermal extremes, indicating that physical priors and ensemble structure can support gridded environmental forecasting beyond wind-speed prediction [33]. Together, these hybrid directions improve accuracy in many studies, yet decomposition-heavy pipelines add tuning degrees of freedom and can reduce portability across locations, while monolithic deep backbones still impose one dominant computation pattern across forecast steps.

Recent forecasting studies increasingly use Mixture-of-Experts (MoE) routing to move beyond fixed ensemble averaging and single-model selection. In wind-speed forecasting, MoE-DEWSF uses a dual-layer dynamic ensemble in which a gating network selects context-aware expert knowledge and a second correction layer adjusts subsequence weights under fluctuating wind conditions [34]. In wind-power forecasting, physics-aware dynamic-graph MoE models blend physics-informed and data-driven predictors through a gate that changes with operating conditions [35]. Related energy-forecasting studies also use MoE modules for multitask horizon sharing and conditional computation inside Transformer blocks, where learned routing allocates model

capacity according to the input context [36], [37]. However, these studies mainly emphasize decomposition-level ensemble correction, physics/data-driven expert blending, multitask output sharing, or internal Transformer capacity allocation. They do not directly isolate horizon-conditioned inference-time routing among explicit persistence, dilated temporal, and locality-preserving recurrent experts for short-term wind-speed prediction from a fixed spatio-temporal mesoscale patch. This gap motivates the proposed Horizon-Conditioned Regime Gate, which assigns expert weights separately at each forecast horizon according to the recent input regime.

Adjacent wind-energy forecasting studies also engage graph-based spatio-temporal learning, uncertainty-aware prediction, and turbine-level operational modeling. Sparse dynamic graph models capture local–global spatial coupling among wind sites for spatio-temporal wind-speed forecasting [38]. Probabilistic wind-speed and wind-power forecasting further quantifies predictive uncertainty through interval, density, Bayesian, and ensemble formulations, which supports risk-aware operation rather than only point-error minimization [39], [40]. In parallel, turbine power-curve modeling links wind conditions to power output and provides an operational bridge between wind forecasting and turbine- or farm-level decision support [41]. These directions complement the present deterministic short-term WSP setting, where the objective is to isolate horizon-conditioned routing under a fixed NASA POWER mesoscale benchmark.

Together, these developments narrow the remaining gap in short-term WSP. Performance improves when a model combines persistence, multi-scale temporal structure, and local spatial interactions, yet many pipelines still implement this combination through a single shared backbone or through step-invariant fusion that applies one computation pattern across all forecast steps. Optimization-centric hybrids strengthen individual components, but they often concentrate adaptivity in offline search and then keep inference-time computation fixed, which can reduce robustness when regimes change. For example, dual-path frameworks extract long-term and fluctuation features and concatenate them without adaptive fusion, while using MoE primarily to optimize hyperparameters rather than to route forecasts at inference [42]. In parallel, compact hybrids target deployability by tuning a single lightweight backbone, but they still execute one shared mapping across forecast steps [43]. This step-dependent and regime-dependent mismatch motivates an inference-time routing mechanism that reallocates capacity as predictability and dominant dynamics evolve across steps. Accordingly, this work formulates WSP as a regime-conditioned fusion problem and learns step-specific mixtures of complementary spatio-temporal experts under a single trainable gate.

This work makes three contributions at the level of problem formulation, architectural synthesis, and empirical validation. First, it formulates short-term WSP as a regime-conditioned, step-aware fusion problem and combines three complementary spatio-temporal experts aligned

with persistence, dilated multi-scale temporal structure, and locality-preserving recurrent dynamics. Second, it develops a Horizon-Conditioned Regime Gate (HCRG) that predicts horizon-specific mixture weights from the input window and performs inference-time fusion rather than static ensembling or offline best-expert selection. Third, it provides an operationally grounded evaluation that isolates step-conditioned routing against strong aggregation baselines and connects wind-speed errors to power and energy proxies, including ramp-region analysis where the power curve is most sensitive to wind-speed deviations.

## II. PROPOSED METHODOLOGY

This work targets multi-horizon wind forecasting from a local spatio-temporal patch by combining complementary inductive biases through a horizon-aware gating mechanism. The system consists of three spatio-temporal experts and a trainable gate. The experts include a recurrent expert that models temporal persistence from a flattened patch, a temporal dilated convolution expert that models multi-scale temporal dependencies with causal dilations, and a convolution recurrent expert that preserves explicit  $3 \times 3$  locality in its recurrent state. The gate, called the Horizon-Conditioned Regime Gate (HCRG), produces horizon-specific mixture weights conditioned on the input window and fuses expert predictions into a single horizon vector. This modular decomposition isolates representation learning inside experts from regime selection in the gate and enables horizon-specialized fusion.

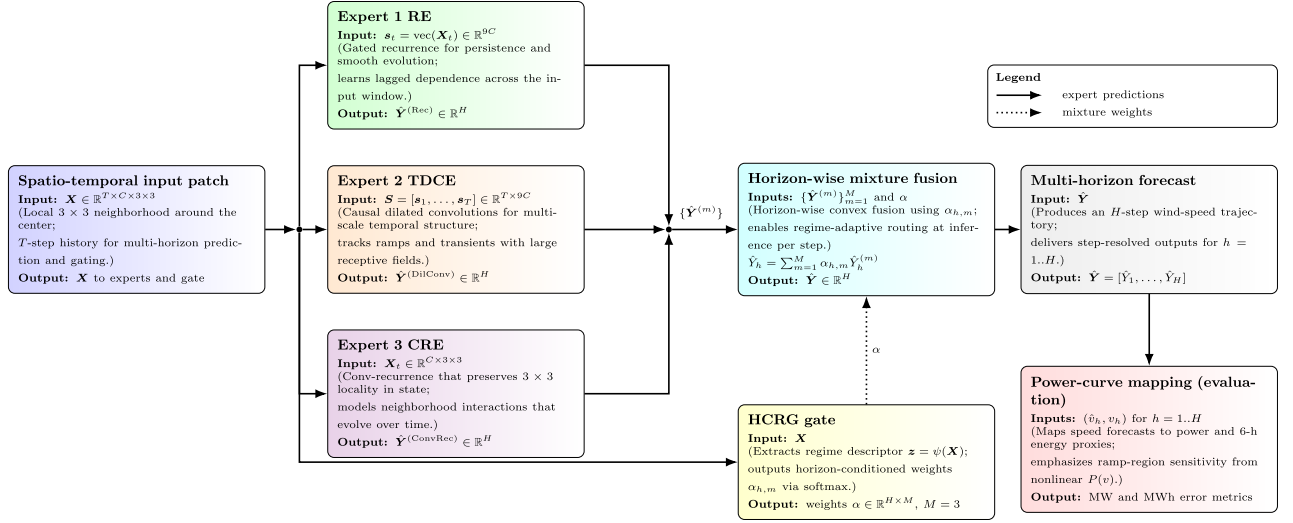
The prediction task maps a recent spatio-temporal history to a vector of  $H$  future targets. The input is

$$\mathbf{X} \in \mathbb{R}^{T \times C \times 3 \times 3}, \quad (2)$$

where  $T$  denotes the lookback length in time steps,  $C$  denotes the number of channels per grid cell, and the  $3 \times 3$  patch provides neighborhood context around the central point. The target is

$$\mathbf{Y} \in \mathbb{R}^H, \quad (3)$$

where  $H$  denotes the forecast horizon count and  $Y_h$  denotes the target at horizon index  $h \in \{1, \dots, H\}$ . An expert predictor is written as  $\hat{\mathbf{Y}}^{(m)} = f_m(\mathbf{X}; \boldsymbol{\theta}_m)$ , where  $m \in \{1, \dots, M\}$  indexes experts and  $\boldsymbol{\theta}_m$  denotes expert parameters. The block diagram in Fig. 1 can be interpreted as a left-to-right workflow. The input patch  $\mathbf{X}$  is first processed by three parallel experts that represent different inductive biases: RE summarizes temporal persistence after flattening the local patch, TDCE extracts causal multi-scale temporal patterns through dilated convolutions, and CRE keeps the  $3 \times 3$  neighborhood structure inside a recurrent state. HCRG then uses the recent regime information to assign horizon-specific weights to the three expert forecasts. Therefore, the expert modules generate candidate forecasts, while the gate decides how much each candidate should contribute at each forecast step.



**FIGURE 1.** Workflow of the proposed Regime-Conditioned Horizon-aware Mixture-of-Experts (RCHA-MOE). The same  $3 \times 3$  spatio-temporal input patch is processed by three complementary experts that emphasize temporal persistence, dilated multi-scale temporal structure, and locality-preserving recurrent dynamics. HCRG then assigns horizon-specific mixture weights and fuses the expert forecasts into the final multi-step prediction.

### A. RECURRENT EXPERT

This expert uses gated recurrence to represent temporal persistence and delayed correlations in the input window while treating the local patch as a vector-valued time series. The  $3 \times 3$  patch is flattened at each time step into

$$s_t = \text{vec}(X_{t,:,:}) \in \mathbb{R}^{9C}, \quad (4)$$

where  $\text{vec}(\cdot)$  stacks spatial entries and channels, and  $t \in \{1, \dots, T\}$  indexes time. The recurrent update maintains a hidden state  $h_t \in \mathbb{R}^d$  and a cell state  $c_t \in \mathbb{R}^d$  through

$$i_t = \sigma(W_i s_t + U_i h_{t-1} + b_i), \quad (5)$$

$$f_t = \sigma(W_f s_t + U_f h_{t-1} + b_f), \quad (6)$$

$$o_t = \sigma(W_o s_t + U_o h_{t-1} + b_o), \quad (7)$$

$$\tilde{c}_t = \tanh(W_c s_t + U_c h_{t-1} + b_c), \quad (8)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad (9)$$

$$h_t = o_t \odot \tanh(c_t), \quad (10)$$

where  $\sigma(\cdot)$  denotes the sigmoid,  $\tanh(\cdot)$  denotes the hyperbolic tangent, and  $\odot$  denotes element-wise multiplication. A linear readout maps the final hidden state to the horizon vector,

$$\hat{Y}^{(\text{Rec})} = W_y h_T + b_y, \quad (11)$$

where  $W_y \in \mathbb{R}^{H \times d}$  and  $b_y \in \mathbb{R}^H$ . This expert learns smooth persistence, lagged dependencies, and regime-dependent temporal memory in  $X$ , but it does not explicitly enforce spatial locality because spatial structure enters only through  $\text{vec}(\cdot)$ .

### B. TEMPORAL DILATED CONVOLUTION EXPERT

This expert uses causal dilated convolutions with residual connections to learn multi-scale temporal structure with an explicit receptive-field schedule. Let  $S \in \mathbb{R}^{T \times 9C}$  stack  $s_t$  over time. A dilated causal convolution produces

$$z_t = \sum_{k=0}^{K-1} W_k s_{t-dk}, \quad (12)$$

where  $K$  denotes the kernel size,  $d$  denotes the dilation factor,  $W_k$  denotes the learnable weight matrix for lag index  $k$ , and causal padding ensures  $z_t$  depends only on  $\{s_\tau\}_{\tau \leq t}$ . Stacking residual blocks with increasing  $d$  expands the receptive field efficiently while preserving gradient flow. The expert prediction is written as

$$\hat{Y}^{(\text{DilConv})} = g_{\text{DilConv}}(S; \theta_{\text{DilConv}}), \quad (13)$$

where  $g_{\text{DilConv}}(\cdot)$  denotes the temporal dilated convolution mapping and  $\theta_{\text{DilConv}}$  denotes its parameters. This expert learns sparse, scale-separated temporal motifs in  $X$ , including abrupt changes and long-memory components, because dilation imposes a structured temporal basis. In contrast to the recurrent expert, which learns memory through gated state updates, this expert learns memory through an explicit receptive field that can represent long lags with fewer sequential dependencies.

### C. CONVOLUTION RECURRENT EXPERT

This expert uses convolutional gating to keep the recurrent state defined on the  $3 \times 3$  patch, which preserves explicit locality and enables neighborhood interactions to shape temporal evolution. At time step  $t$ , the input slice is



$X_t \in \mathbb{R}^{C \times 3 \times 3}$ , the hidden state is  $H_t \in \mathbb{R}^{D \times 3 \times 3}$ , and the cell state is  $C_t \in \mathbb{R}^{D \times 3 \times 3}$ . The update equations are

$$I_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + b_i), \quad (14)$$

$$F_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + b_f), \quad (15)$$

$$O_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + b_o), \quad (16)$$

$$\tilde{C}_t = \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c), \quad (17)$$

$$C_t = F_t \odot C_{t-1} + I_t \odot \tilde{C}_t, \quad (18)$$

$$H_t = O_t \odot \tanh(C_t), \quad (19)$$

where  $*$  denotes a 2D convolution over the  $3 \times 3$  grid and  $W_x$  and  $W_h$  denote convolution kernels applied to the input and hidden state. A readout maps the final hidden tensor to the horizon vector,

$$\hat{Y}^{(\text{ConvRec})} = g_{\text{ConvRec}}(H_T; \theta_{\text{ConvRec}}), \quad (20)$$

where  $g_{\text{ConvRec}}(\cdot)$  denotes the readout and  $\theta_{\text{ConvRec}}$  denotes its parameters. This expert learns coupled spatio-temporal patterns in  $X$ , such as neighborhood gradients and coherent local structures that propagate over time. In contrast to the previous two experts, spatial locality is preserved inside the recurrent state, so the learned dynamics incorporate both temporal memory and local spatial interactions.

#### D. HORIZON-CONDITIONED REGIME GATE

The Horizon-Conditioned Regime Gate (HCRG) learns horizon-specific mixture weights from the shared context representation. In this work, a regime denotes a recurrent local dynamical state of the forecasting process, identified from the recent joint behavior of the input sequence, under which the relative predictive utility of the experts changes across forecast horizons. A regime descriptor is computed as

$$z = \psi(X) \in \mathbb{R}^{d_z}, \quad (21)$$

where  $\psi(\cdot)$  denotes a feature extractor and  $d_z$  denotes the descriptor dimension. Horizon-wise logits follow as

$$\ell_{h,m} = w_{h,m}^\top z + b_{h,m}, \quad (22)$$

where  $w_{h,m} \in \mathbb{R}^{d_z}$  and  $b_{h,m} \in \mathbb{R}$  parameterize the gate for horizon  $h$  and expert  $m$ . Mixture weights are defined by

$$\alpha_{h,m} = \frac{\exp(\ell_{h,m})}{\sum_{j=1}^M \exp(\ell_{h,j})}, \quad (23)$$

where  $M$  denotes the number of experts and  $\sum_{m=1}^M \alpha_{h,m} = 1$  for each horizon  $h$ . The fused prediction is computed horizon-wise as

$$\hat{Y}_h = \sum_{m=1}^M \alpha_{h,m} \hat{Y}_h^{(m)}, \quad (24)$$

where  $\hat{Y}_h^{(m)}$  denotes the  $h$ th component of expert  $m$ 's output. This gate learns a regime-to-expert assignment policy from  $X$  and learns horizon specialization because  $\alpha_{h,m}$  varies with  $h$ . Operationally, the learned mixture weights provide a practical interpretation of the local forecasting regime.

A larger RE weight indicates that the recent wind trajectory is sufficiently persistent for recurrent memory to dominate. A larger TDCE weight indicates that the input window contains scale-separated temporal changes that are better represented through dilated causal filters. A larger CRE weight indicates that neighborhood structure in the  $3 \times 3$  patch contributes more strongly to the forecast. The regimes are therefore interpreted as input-dependent predictive states rather than as manually labeled meteorological classes, which avoids imposing external labels that are not directly observed in the NASA POWER benchmark.

#### E. TWO-STAGE TRAINING OBJECTIVE

Training proceeds in two stages. First, each expert is trained independently to convergence using a horizon-averaged mean-squared error. For a dataset of  $N$  samples, the expert objective is

$$\mathcal{L}_{\text{exp}}(\theta_m) = \frac{1}{N} \sum_{n=1}^N \frac{1}{H} \sum_{h=1}^H \left( \hat{Y}_{n,h}^{(m)} - Y_{n,h} \right)^2, \quad (25)$$

where  $Y_{n,h}$  denotes the ground-truth target for sample  $n$  at horizon  $h$  and  $\hat{Y}_{n,h}^{(m)}$  denotes the corresponding prediction from expert  $m$ . This objective forces each expert to explain the full horizon vector, so each expert learns horizon-dependent error structure induced by  $X$ .

Second, the gate is optimized after expert training by minimizing the multi-horizon regression error of the fused predictor while keeping expert predictors frozen. The primary objective is

$$\mathcal{L}_{\text{gate}}(\phi) = \frac{1}{N} \sum_{n=1}^N \frac{1}{H} \sum_{h=1}^H \left( \hat{Y}_{n,h} - Y_{n,h} \right)^2, \quad (26)$$

where  $\phi$  denotes gate parameters,  $\hat{Y}_{n,h}$  denotes the fused prediction, and  $Y_{n,h}$  denotes ground truth. The optional entropy regularizer is

$$\mathcal{R}(\phi) = \frac{1}{N} \sum_{n=1}^N \frac{1}{H} \sum_{h=1}^H \left( - \sum_{m=1}^M \alpha_{n,h,m} \log \alpha_{n,h,m} \right), \quad (27)$$

where  $\alpha_{n,h,m}$  denotes the gate weight for sample  $n$ , horizon  $h$ , and expert  $m$ . The final objective is

$$\min_{\phi} \mathcal{L}_{\text{gate}}(\phi) + \lambda \mathcal{R}(\phi), \quad (28)$$

where  $\lambda \geq 0$  controls the regularization strength. Together, this two-stage optimization first learns horizon-aware expert predictors and then learns regime-conditioned horizon-specific fusion that reduces multi-horizon error. Algorithm 1 summarizes the complete training and inference procedure.

### III. RESULTS AND COMPARISON

This section reports empirical evidence for the proposed Regime-Conditioned Horizon-aware Mixture-of-Experts (RCHA-MOE) under a controlled multi-horizon

**Algorithm 1** Two-Stage Training and Inference of RCHA-MOE

**Require:**  $\mathcal{D} = \{(X_n, Y_n)\}_{n=1}^N$ , experts  $\{f_m(\cdot; \theta_m)\}_{m=1}^M$  with  $M = 3$ , gate feature extractor  $\psi(\cdot; \phi)$ , horizons  $H$ , entropy weight  $\lambda$

**Require:** Learning rates  $\eta_{\text{exp}}$ ,  $\eta_{\text{gate}}$ , epochs  $E_{\text{exp}}$ ,  $E_{\text{gate}}$ , batch size  $B$

**Ensure:** Trained experts  $\{\theta_m\}$  and gate  $\phi$

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1: Initialize  $\{\theta_m\}_{m=1}^M$  and  $\phi$ 
2: Stage 1 Train experts independently
3: for  $m = 1$  to  $M$  do
4:   for  $e = 1$  to  $E_{\text{exp}}$  do
5:     Sample minibatch  $\mathcal{B}$  with  $|\mathcal{B}| = B$ 
6:      $\mathcal{L}_{\text{exp}} \leftarrow \frac{1}{B} \sum_{(X,Y) \in \mathcal{B}} \frac{1}{H} \sum_{h=1}^H (f_m(X; \theta_m)_h - Y_h)^2$ 
7:      $\theta_m \leftarrow \theta_m - \eta_{\text{exp}} \nabla_{\theta_m} \mathcal{L}_{\text{exp}}$ 
8:   end for
9:   Freeze  $\theta_m$ 
10: end for
11: Stage 2 Train gate with frozen experts
12: for  $e = 1$  to  $E_{\text{gate}}$  do
13:   Sample minibatch  $\mathcal{B}$  with  $|\mathcal{B}| = B$ 
14:    $\mathcal{L}_{\text{gate}} \leftarrow 0$ 
15:   for all  $(X, Y) \in \mathcal{B}$  do
16:      $\mathbf{z} \leftarrow \psi(X; \phi)$  and  $\hat{Y}^{(m)} \leftarrow f_m(X; \theta_m)$  for all  $m$ 
17:     for  $h = 1$  to  $H$  do
18:        $\alpha_{h,1:M} \leftarrow \text{softmax}(\{\mathbf{w}_{h,m}^\top \mathbf{z} + b_{h,m}\}_{m=1}^M)$ 
19:        $\hat{Y}_h \leftarrow \sum_{m=1}^M \alpha_{h,m} \hat{Y}_h^{(m)}$ 
20:     end for
21:      $\mathcal{L}_{\text{mse}} \leftarrow \frac{1}{H} \sum_{h=1}^H (\hat{Y}_h - Y_h)^2$ 
22:      $\mathcal{L}_{\text{ent}} \leftarrow \frac{1}{H} \sum_{h=1}^H \left( - \sum_{m=1}^M \alpha_{h,m} \log \alpha_{h,m} \right)$ 
23:      $\mathcal{L}_{\text{gate}} \leftarrow \mathcal{L}_{\text{gate}} + \mathcal{L}_{\text{mse}} + \lambda \mathcal{L}_{\text{ent}}$ 
24:   end for
25:    $\mathcal{L}_{\text{gate}} \leftarrow \mathcal{L}_{\text{gate}} / B$ 
26:    $\phi \leftarrow \phi - \eta_{\text{gate}} \nabla_{\phi} \mathcal{L}_{\text{gate}}$ 
27: end for
28: Inference for input  $X$ 
29:  $\mathbf{z} \leftarrow \psi(X; \phi)$  and  $\hat{Y}^{(m)} \leftarrow f_m(X; \theta_m)$  for all  $m$ 
30: for  $h = 1$  to  $H$  do
31:    $\alpha_{h,1:M} \leftarrow \text{softmax}(\{\mathbf{w}_{h,m}^\top \mathbf{z} + b_{h,m}\}_{m=1}^M)$ 
32:   Output  $\hat{Y}_h \leftarrow \sum_{m=1}^M \alpha_{h,m} \hat{Y}_h^{(m)}$ 
33: end for

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forecasting protocol. The evaluation fixes the sampling rate to 1 hour, the input window to  $T = 24$  hours, and the forecast horizon to  $H = 6$  hours, and it uses identical chronological splits and a shared hyperparameter tuning budget across all baselines to isolate architectural effects. This section first describes the datasets and their seasonal structure, then specifies the experimental setup and evaluation metrics, and finally compares RCHA-MOE against state-of-the-art baselines through ablations, horizon-wise analyses, and downstream power-curve proxies that translate wind-speed errors into operational impact.

**A. DATASET**

This study evaluates the proposed RCHA-MOE on three hourly, location-specific wind datasets retrieved from NASA's Prediction Of Worldwide Energy Resources (POWER) project [44]. Each dataset corresponds to one city-center coordinate together with an associated  $3 \times 3$  neighborhood patch, which defines a reproducible spatio-temporal field under a fixed sampling rule. Specifically, this study queries POWER at the city-center coordinates for Lerwick (UK), Tarifa (Spain), and Wellington (New Zealand), and it augments each center point with eight surrounding points (N, NE, E, SE, S, SW, W, NW) placed at an approximate 100-km offset along the corresponding bearings. At each grid point and hour, POWER provides wind speed at 50 m above ground level (m/s), wind direction at 50 m (deg, clockwise from north), surface pressure (kPa), specific humidity at 2 m (g/kg), and air temperature at 2 m ( $^{\circ}\text{C}$ ). All three datasets span 2015-01-01 00:00 to 2024-12-31 23:00 at a 1-h sampling rate, yielding 87,672 hours per location. This study splits each time series chronologically into 70% training, 10% validation, and 20% testing, and Table 1 reports the resulting hour counts together with central-point wind-speed statistics. Figure 2 summarizes seasonal structure using monthly statistics. Lerwick shows the largest seasonal amplitude, with monthly mean  $U_{50}$  ranging from 7.26 m/s (Jul) to 12.25 m/s (Jan), and the largest dispersion, with monthly  $\sigma(U_{50})$  peaking at 5.34 m/s (Dec). In contrast, Tarifa exhibits a smaller seasonal swing, with monthly mean  $U_{50}$  spanning 5.46–6.99 m/s and peak  $\sigma(U_{50})$  of 3.59 m/s (Mar). Moreover, Wellington shows a Southern Hemisphere seasonal phase, with monthly mean  $U_{50}$  spanning 7.85 m/s (Mar) to 9.38 m/s (Sep) and peak  $\sigma(U_{50})$  of 4.25 m/s (Jul). Together, the  $3 \times 3$  patch and the multi-season regimes define a controlled setting for evaluating multi-horizon spatio-temporal wind-speed forecasting under a consistent data product and sampling rule.

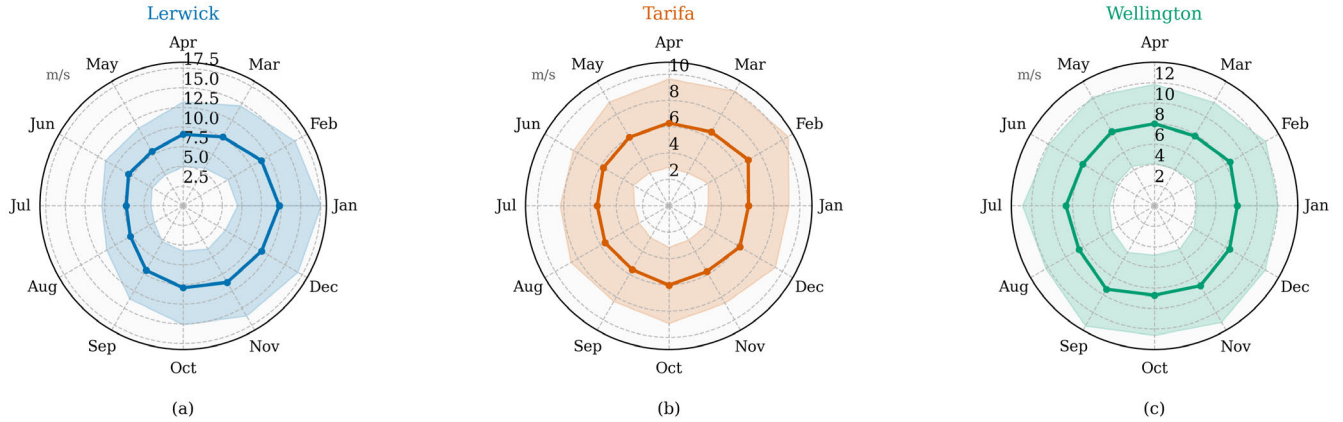
**B. EXPERIMENTAL SETUP**

To ensure a controlled and reproducible comparison between the proposed method and established baselines, this study standardizes the forecasting task, the sample-construction pipeline, and the model-selection protocol across all methods. Each supervised example uses the previous  $T = 24$  consecutive hours of spatio-temporal observations from the  $3 \times 3$  patch as input and predicts the next  $H = 6$  hours of wind speed at the central point as output. Within each split, this study constructs samples with a sliding window and a stride of 1 hour, so adjacent examples overlap by  $T - 1$  hours and differ only in the newly appended hour. Because each example consumes  $T$  input hours and requires  $H$  future labels, a split with  $L$  hours yields  $N = L - (T + H) + 1$  examples. Using the split sizes in Table 1, this protocol produces 61,341 training examples, 8,738 validation examples, and 17,506 test examples per dataset. Table 2 summarizes the common experimental configuration.

**TABLE 1.** Dataset coordinates, split sizes, and central-point wind-speed statistics ( $U_{50}$  at C).

| Dataset    | Lat (°)   | Lon (°)   | Period                | Total (h) | Train (h) | Val (h) | Test (h) | $\mu(U_{50})$ (m/s) | $\sigma(U_{50})$ (m/s) | $\min(U_{50})$ (m/s) | $\max(U_{50})$ (m/s) |
|------------|-----------|-----------|-----------------------|-----------|-----------|---------|----------|---------------------|------------------------|----------------------|----------------------|
| Lerwick    | 60.15340  | -1.14427  | 2015-01-01–2024-12-31 | 87672     | 61370     | 8767    | 17535    | 9.7078              | 4.6570                 | 0.04                 | 40.25                |
| Tarifa     | 36.01393  | -5.60695  | 2015-01-01–2024-12-31 | 87672     | 61370     | 8767    | 17535    | 6.0236              | 3.1008                 | 0.03                 | 19.93                |
| Wellington | -41.29477 | 174.77197 | 2015-01-01–2024-12-31 | 87672     | 61370     | 8767    | 17535    | 8.4664              | 3.9818                 | 0.05                 | 25.59                |

Monthly wind speed seasonal wheel (mean  $\pm$  1 std)



**FIGURE 2.** Monthly seasonal wheel of central-point wind speed ( $U_{50}$  at C) for Lerwick, Tarifa, and Wellington. The radial curve reports the monthly mean, and the shaded band shows mean  $\pm$  1 standard deviation over 2015–2024, highlighting differences in seasonal amplitude and dispersion across the three sites.

**TABLE 2.** Experimental configuration used across all methods.

| Setting                   | Value              |
|---------------------------|--------------------|
| Sampling rate             | 1 hour             |
| Input length $T$          | 24                 |
| Forecast horizon $H$      | 6                  |
| Sliding stride            | 1 hour             |
| Optimizer (neural models) | Adam               |
| Max epochs                | 1000               |
| Early stopping patience   | 10                 |
| Training batch size       | 128                |
| Evaluation batch size     | 512                |
| Default learning rate     | $10^{-3}$          |
| Hardware                  | NVIDIA RTX 3080 Ti |

All neural models use the Adam optimizer, train for up to 1000 epochs, and apply early stopping with patience 10 based on validation loss. This study fixes the training batch size to 128 and the evaluation batch size to 512 across methods, with a default learning rate of  $10^{-3}$ . For each method, this study optimizes hyperparameters via Tree-structured Parzen Estimation (TPE) on a 5% tuning subset sampled from the training split, retrain the method using the selected configuration, and reports performance on the held-out test subset. This study uses the same tuning budget for all methods to preserve a same-budget comparison under a

shared protocol, which can be conservative for higher-capacity architectures that often benefit from larger search budgets. Concretely, the TPE space covers each model’s primary capacity and regularization knobs. For iTransformer and PatchTST, the search includes depth, model dimension, number of heads, patch-related tokenization choices, and dropout in addition to learning-rate-related terms. For tree and randomized-feature baselines, the search includes the main complexity controls such as tree depth and number of estimators or random-node count together with readout regularization.

For reproducibility, the final RCHA-MOE implementation uses three fixed expert families and a separately trained HCRG gate. RE receives the flattened  $3 \times 3$  patch at each time step, maps the 9C-dimensional input sequence through two recurrent layers with 128 hidden units, and predicts the  $H = 6$  future steps with GELU activation. TDCE projects the flattened 9C-dimensional sequence to 128 channels and applies five causal temporal blocks with kernel size 3 and dilation factors  $\{1, 2, 4, 8, 16\}$ , using GELU activation. CRE embeds each local  $3 \times 3$  input slice into 16 channels, applies two recurrent convolutional cells with  $3 \times 3$  kernels, and predicts from the final center-cell hidden representation with GELU activation. The HCRG gate uses a normalized regime descriptor  $\mathbf{z} \in \mathbb{R}^{d_z}$  with  $d_z = 312$ , computed from input-window statistics, center-point history, and local spatial-gradient and Laplacian terms. Inputs use wind speed at 50 m, wind direction at 50 m, surface pressure,

**TABLE 3.** Architecture details, HCRG search space, and selected site-specific settings.

| Component                                                | Units/channels     | Depth/blocks                    | Kernel/dilation                 | Activation            | Setting                     |
|----------------------------------------------------------|--------------------|---------------------------------|---------------------------------|-----------------------|-----------------------------|
| RE                                                       | 128 hidden units   | 2 recurrent layers              | –                               | GELU                  | fixed                       |
| TDCE                                                     | 128 channels       | 5 causal temporal blocks        | $k = 3, d = \{1, 2, 4, 8, 16\}$ | GELU                  | fixed                       |
| CRE                                                      | 16 hidden channels | 2 recurrent convolutional cells | $3 \times 3$ kernels            | GELU                  | fixed                       |
| HCRG descriptor                                          | $d_z = 312$        | –                               | –                               | –                     | fixed normalized descriptor |
| <i>HCRG searched hyperparameters and selected values</i> |                    |                                 |                                 |                       |                             |
| Symbol                                                   | Description        | Search space                    | Lerwick                         | Tarifa                | Wellington                  |
| $d_g$                                                    | Hidden width       | $\{128, 192, 256, 384, 512\}$   | 384                             | 512                   | 128                         |
| $L_g$                                                    | Gate depth         | $\{2, 3, 4\}$                   | 2                               | 3                     | 2                           |
| $p_g$                                                    | Dropout            | $[0, 0.35]$                     | 0.1192                          | 0.2629                | 0.2522                      |
| $\lambda$                                                | Entropy weight     | $10^{-5}$ – $10^{-2}$           | $2.30 \times 10^{-3}$           | $6.81 \times 10^{-3}$ | $9.36 \times 10^{-5}$       |

specific humidity at 2 m, and air temperature at 2 m on the  $3 \times 3$  grid, with central-point wind speed at 50 m as the target. Wind direction is converted to sine and cosine components, and predictors, targets, and gate descriptors are normalized using training-split z-score statistics only. All neural components use seed 123, Adam optimization, maximum 1000 epochs, early-stopping patience 10, training batch size 128, evaluation batch size 512, and validation performance for model selection. During inference, the trained experts produce  $M \times H$  predictions, the stored descriptor statistics normalize the HCRG input, the saved HCRG checkpoint produces horizon-wise mixture weights, and the final prediction is de-normalized using stored target statistics. The HCRG search used 24 validation-driven configurations under the shared tuning protocol. Table 3 reports the expert units, kernel and dilation settings, activation choices, and HCRG search space with selected site-specific settings.

### C. EVALUATION METRICS

This study evaluates multi-horizon point forecasts using root mean-squared error (RMSE), mean absolute error (MAE), and coefficient of determination ( $R^2$ ). Let  $Y_{n,h}$  denote the ground-truth wind speed (m/s) for test sample  $n \in \{1, \dots, N\}$  at horizon  $h \in \{1, \dots, H\}$ , and let  $\hat{Y}_{n,h}$  denote the corresponding prediction. This study aggregates errors over all horizons and all test samples to report a single metric per model and dataset.

RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{NH} \sum_{n=1}^N \sum_{h=1}^H (\hat{Y}_{n,h} - Y_{n,h})^2}. \quad (29)$$

RMSE emphasizes larger deviations because it squares residuals before averaging, which often makes it sensitive to large wind-speed errors at specific horizons.

MAE is defined as

$$\text{MAE} = \frac{1}{NH} \sum_{n=1}^N \sum_{h=1}^H |\hat{Y}_{n,h} - Y_{n,h}|. \quad (30)$$

MAE measures typical absolute deviation in m/s and often provides a robust complement to RMSE when outliers occur.

The coefficient of determination is computed as

$$R^2 = 1 - \frac{\sum_{n=1}^N \sum_{h=1}^H (\hat{Y}_{n,h} - Y_{n,h})^2}{\sum_{n=1}^N \sum_{h=1}^H (Y_{n,h} - \bar{Y})^2}, \quad (31)$$

$$\bar{Y} = \frac{1}{NH} \sum_{n=1}^N \sum_{h=1}^H Y_{n,h}. \quad (32)$$

$R^2$  reports the fraction of variance explained relative to a constant predictor that always outputs the test-set mean  $\bar{Y}$  aggregated over horizons.

### D. ABLATION STUDY

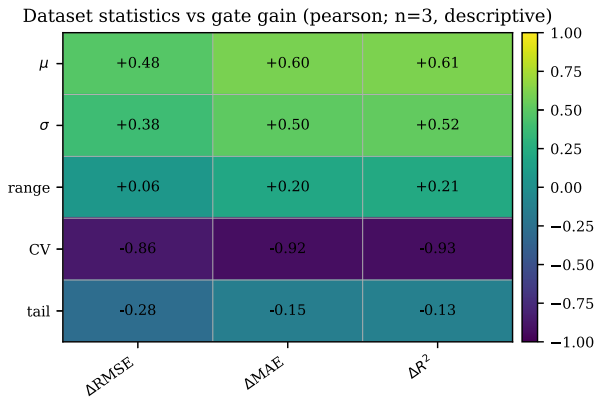
This study decomposes the proposed Regime-Conditioned Horizon-aware Mixture-of-Experts (RCHA-MOE) into its constituent components to attribute performance gains. The ablation compares three standalone experts, each encoding a distinct inductive bias, against two aggregation baselines and the full gated mixture. The experts include a Recurrent Expert (RE) that captures temporal persistence, a Temporal Dilated Convolution Expert (TDCE) that captures causal multi-scale temporal structure through dilations, and a Convolutional Recurrent Expert (CRE) that preserves explicit  $3 \times 3$  spatial locality while modeling temporal dynamics. In addition, this study evaluates a Uniform Average (UA) ensemble that assigns equal weights to all experts and a Best Expert (BE) selector that chooses, for each dataset, the single expert that minimizes validation RMSE. This study then reports that expert's test RMSE, MAE, and  $R^2$  without re-optimizing per metric. Finally, RCHA-MOE evaluates whether horizon-conditioned gating yields gains beyond uniform averaging by adapting mixture weights across horizons and input regimes.

Table 4 shows that the learned mixture improves over any single expert across all datasets, which indicates that complementary inductive biases cover regimes that a single expert misses. BE tracks the strongest standalone expert and consistently underperforms UA and RCHA-MOE, which suggests that no single expert dominates across time



**TABLE 4.** Ablation mean test metrics across datasets.

| Dataset    | Variant  | RMSE          | MAE           | $R^2$         |
|------------|----------|---------------|---------------|---------------|
| Lerwick    | RE       | 1.0487        | 0.7452        | 0.9387        |
|            | TDCE     | 1.0473        | 0.7509        | 0.9382        |
|            | CRE      | 1.0436        | 0.7471        | 0.9400        |
|            | RCHA-MOE | <b>0.9777</b> | <b>0.6908</b> | 0.9450        |
|            | UA       | 0.9802        | 0.6953        | <b>0.9453</b> |
|            | BE       | 1.0473        | 0.7509        | 0.9382        |
| Tarifa     | RE       | 0.8040        | 0.5981        | 0.9205        |
|            | TDCE     | 0.8314        | 0.6196        | 0.9151        |
|            | CRE      | 0.8122        | 0.6043        | 0.9190        |
|            | RCHA-MOE | 0.7675        | <b>0.5682</b> | 0.9260        |
|            | UA       | <b>0.7660</b> | 0.5692        | <b>0.9272</b> |
|            | BE       | 0.8122        | 0.6043        | 0.9190        |
| Wellington | RE       | 1.0373        | 0.7612        | 0.9157        |
|            | TDCE     | 1.0270        | 0.7617        | 0.9176        |
|            | CRE      | 1.0229        | 0.7513        | 0.9185        |
|            | RCHA-MOE | <b>0.9573</b> | <b>0.7003</b> | <b>0.9261</b> |
|            | UA       | 0.9686        | 0.7093        | 0.9253        |
|            | BE       | 1.0270        | 0.7617        | 0.9176        |

**FIGURE 3.** Descriptive Pearson correlation between dataset wind statistics and gate gains over uniform averaging across datasets ( $n = 3$ ). Gate gains are  $\Delta RMSE = RMSE_{ua} - RMSE_{gate}$ ,  $\Delta MAE = MAE_{ua} - MAE_{gate}$ , and  $\Delta R^2 = R^2_{gate} - R^2_{ua}$ , where positive values indicate that gating improves upon uniform averaging.

and horizons. Moreover, RCHA-MOE matches or improves upon UA on two datasets. Wellington improves on all three metrics (RMSE 0.9573 vs. 0.9686, MAE 0.7003 vs. 0.7093,  $R^2$  0.9261 vs. 0.9253), while Lerwick improves RMSE and MAE (0.9777 vs. 0.9802 and 0.6908 vs. 0.6953) with a negligible  $R^2$  trade-off (0.9450 vs. 0.9453). In contrast, Tarifa becomes a near-tie where UA attains slightly better RMSE and  $R^2$  (0.7660 and 0.9272) while RCHA-MOE attains slightly better MAE (0.5682). Together, these results position UA as a strong aggregation baseline and motivate a dataset-dependent analysis of when horizon-conditioned gating adds value.

From a deployment perspective, HCRG provides a practical middle ground between static averaging and repeated

best-model selection. UA is simple but applies the same expert weights to all regimes and horizons, while BE selects one expert from validation behavior and cannot adapt when the dominant dynamics change during operation. In contrast, RCHA-MOE trains the experts once and uses a lightweight gate at inference to adjust horizon-wise weights from the current input regime. This design avoids manual switching rules or per-site expert reselection, while retaining the ability to favor persistence-, dilation-, or locality-oriented experts when their predictive utility changes.

To isolate the effect of gating, this study defines gate gain relative to uniform averaging as  $\Delta RMSE = RMSE_{ua} - RMSE_{gate}$ ,  $\Delta MAE = MAE_{ua} - MAE_{gate}$ , and  $\Delta R^2 = R^2_{gate} - R^2_{ua}$ , so positive values indicate that gating improves upon UA. Figure 3 reports Pearson correlations between these gains and per-dataset wind statistics (Table 1) across the three datasets ( $n = 3$ ). The correlations suggest that gate gains tend to increase with absolute wind level and dispersion, while they tend to decrease as relative variability increases when summarized by the coefficient of variation  $CV = \sigma/\mu$ . This pattern is consistent with the near-tie behavior on Tarifa, which exhibits lower absolute dispersion, and the clearer gains on Lerwick and Wellington, which exhibit larger absolute variability where expert preferences can vary more strongly across regimes. Because  $n = 3$  does not support statistical inference, this analysis serves only as a descriptive diagnostic that motivates the regime-separability interpretation of the gate rather than as a general rule.

#### E. COMPARISON WITH STATE-OF-THE-ART METHODS

This study benchmarks the proposed RCHA-MOE against representative baselines that cover transformer-based sequence modeling (iTransformer [27], PatchTST [26]), hybrid spatio-temporal learners (CoReSTL [45], SHM [43]), randomized feature learning with ridge readout (eDRVFL), and a gradient-boosting pipeline (CANet-FTI-CatBoost [42]). To ensure a consistent evaluation protocol, this study trains all methods on identical splits and input–output construction and tunes hyperparameters using TPE under the same tuning budget. Table 5 reports mean test metrics, while Figure 4 summarizes horizon-wise behavior for RMSE, MAE, and  $R^2$ . Because the input context is a compact  $3 \times 3$  patch with  $T = 24$  and  $H = 6$ , these comparisons reflect performance under a constrained same-budget setting, which can be conservative for models that typically benefit from longer contexts or larger tuning budgets. The baseline set is positioned as a machine-learning and hybrid-model comparison under a common multivariate spatio-temporal protocol. In addition to recent deep and hybrid forecasters, eDRVFL provides a lightweight randomized-feature learner, while CANet-FTI-CatBoost provides a tree-based learning comparator. The Transformer baselines are trained with the same chronological splits, input–output construction, hyperparameter-search budget, early-stopping rule, and held-out test protocol as the other models. Their results are

**TABLE 5.** Mean test metrics for RCHA-MOE and competing baselines across the three datasets. The closest competitor and percentage improvement rows report the strongest non-RCHA-MOE baseline for each dataset.

| Dataset            | Model              | RMSE          | MAE           | $R^2$         |
|--------------------|--------------------|---------------|---------------|---------------|
| Lerwick            | RCHA-MOE           | <b>0.9777</b> | <b>0.6908</b> | <b>0.9450</b> |
|                    | eDRVFL             | 1.0928        | 0.7705        | 0.9308        |
|                    | iTransformer       | 2.8451        | 2.1910        | 0.6285        |
|                    | PatchTST           | 2.9445        | 2.2790        | 0.6015        |
|                    | CoReSTL            | 1.3351        | 0.9421        | 0.9144        |
|                    | SHM                | 1.0633        | 0.7593        | 0.9369        |
|                    | CANet-FTI-CatBoost | 1.0492        | 0.7480        | 0.9378        |
| Closest competitor | CANet-FTI-CatBoost | 1.0492        | 0.7480        | 0.9378        |
| % improvement      |                    | +6.81%        | +7.65%        | +0.77%        |
| Tarifa             | RCHA-MOE           | <b>0.7675</b> | <b>0.5682</b> | <b>0.9260</b> |
|                    | eDRVFL             | 0.8721        | 0.6485        | 0.9043        |
|                    | iTransformer       | 1.3550        | 0.9994        | 0.8014        |
|                    | PatchTST           | 1.3568        | 1.0096        | 0.8001        |
|                    | CoReSTL            | 0.9524        | 0.7123        | 0.8995        |
|                    | SHM                | 0.8246        | 0.6153        | 0.9167        |
|                    | CANet-FTI-CatBoost | 0.8353        | 0.6248        | 0.9127        |
| Closest competitor | SHM                | 0.8246        | 0.6153        | 0.9167        |
| % improvement      |                    | +6.93%        | +7.65%        | +1.02%        |
| Wellington         | RCHA-MOE           | <b>0.9573</b> | <b>0.7003</b> | <b>0.9261</b> |
|                    | eDRVFL             | 1.1590        | 0.8477        | 0.8927        |
|                    | iTransformer       | 1.9352        | 1.4830        | 0.7521        |
|                    | PatchTST           | 2.0366        | 1.5555        | 0.7254        |
|                    | CoReSTL            | 1.3486        | 0.9846        | 0.8764        |
|                    | SHM                | 1.0232        | 0.7570        | 0.9172        |
|                    | CANet-FTI-CatBoost | 1.0558        | 0.7712        | 0.9112        |
| Closest competitor | SHM                | 1.0232        | 0.7570        | 0.9172        |
| % improvement      |                    | +6.44%        | +7.49%        | +0.97%        |

therefore interpreted only within the present short-horizon setting with  $H = 6$ , rather than as a general statement about Transformer-based forecasting.

Attention-based Transformers often deliver their largest gains when longer contexts and longer horizons make global token interactions central to error reduction, whereas short-horizon forecasting with compact contexts often emphasizes persistence and step-local transients that favor explicit multi-scale and recurrent inductive biases. Recent empirical analyses also question whether permutation-invariant self-attention preserves temporal order efficiently for forecasting, and they show that simple linear or streamlined alternatives can match or outperform Transformer-based forecasters on standard benchmarks [46]. In parallel, focused studies of Transformer components argue that self-attention can induce temporal information loss for forecasting and motivate architectures that reduce reliance on self-attention while retaining horizon-aware parameterization [47]. Therefore, the weaker iTransformer and PatchTST results in this NASA POWER setting with  $T = 24$  and  $H = 6$  under a same-budget protocol are consistent with the broader observation that attention is not uniformly dominant across regimes and horizons, rather than implying that attention-based forecasting is ineffective in general.

RCHA-MOE achieves the best mean performance on all three datasets and maintains consistent horizon-wise advantages for RMSE and MAE in Figure 4. On Lerwick,

RCHA-MOE improves over the closest competitor (CANet-FTI-CatBoost) by 6.81% in RMSE and 7.65% in MAE while increasing  $R^2$  by 0.77%. On Tarifa, the strongest competitor becomes SHM, and RCHA-MOE improves by 6.93% (RMSE), 7.65% (MAE), and 1.02% ( $R^2$ ). On Wellington, RCHA-MOE again outperforms SHM by 6.44% (RMSE), 7.49% (MAE), and 0.97% ( $R^2$ ). These gains indicate that horizon-conditioned mixture weighting remains beneficial after per-method tuning under a shared protocol.

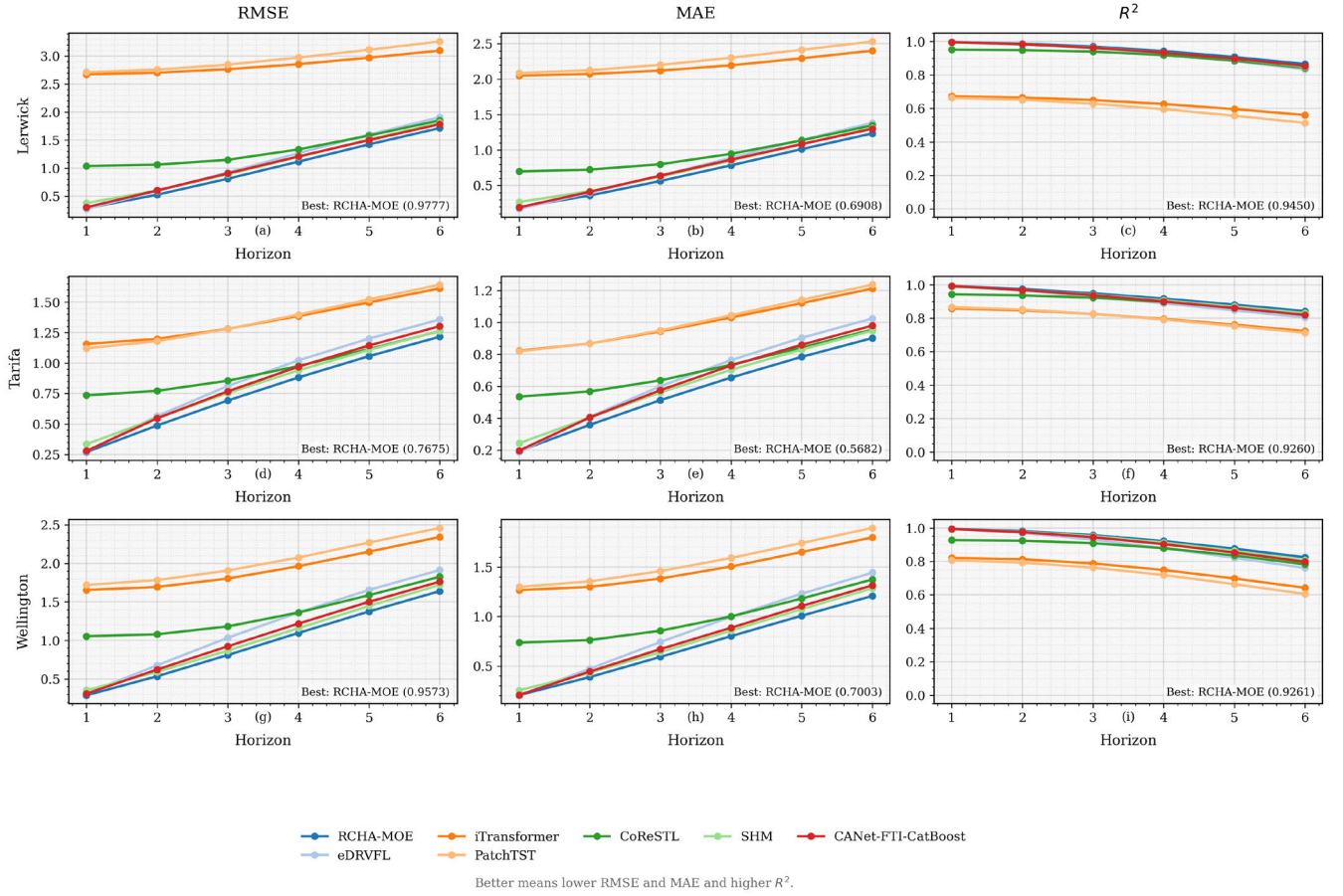
To test whether the observed improvements reflect systematic differences rather than incidental variation, this study applies paired Wilcoxon signed-rank tests between RCHA-MOE and each competing baseline. For statistical testing, this study uses batch-level paired metric estimates rather than individual forecast-target errors. This choice avoids treating highly overlapping sliding-window samples as independent observations. For each dataset and comparator, the 17,506 test windows are divided into batches of 32 consecutive windows, and each batch aggregates  $32 \times 6$  matched forecast targets across the six horizons. RMSE, MAE, and  $R^2$  are recomputed within each batch for RCHA-MOE and the comparator, and a two-sided Wilcoxon signed-rank test is applied to the resulting paired batch-level metric sequences. For  $R^2$ , the sign is inverted before testing so that all metrics follow a lower-is-better convention, and the resulting  $p$ -values are adjusted across baselines using the Benjamini–Hochberg procedure. Figure 5 reports  $-\log_{10}(p_{\text{adj}})$  for each baseline across metrics and datasets, where larger values indicate stronger evidence of a distributional shift in favor of RCHA-MOE.

Across datasets and metrics, the corrected  $p$ -values remain well below common thresholds for most comparisons, which supports the conclusion that RCHA-MOE yields statistically distinguishable improvements relative to the evaluated baselines under the shared protocol. Together, the mean-metric gains (Table 5), the horizon-wise trends (Figure 4), and the nonparametric significance tests (Figure 5) provide convergent evidence that the proposed regime-conditioned, horizon-aware gating improves multi-horizon wind-speed forecasting in this NASA POWER setting.

## F. POWER-CURVE EVALUATION

This study evaluates downstream energy impact by mapping wind-speed forecasts to turbine power using a fixed reference power curve. Let  $v$  (m/s) denote the 50-m wind speed at the central point and let  $\hat{v}$  denote its forecast. The mapping  $P(v)$  (MW) uses cut-in  $v_{\text{ci}} = 3$  m/s, rated speed  $v_r = 11.4$  m/s, cut-out  $v_{\text{co}} = 25$  m/s, and rated power  $P_r = 5$  MW:

$$P(v) = \begin{cases} 0, & v < v_{\text{ci}} \text{ or } v > v_{\text{co}}, \\ P_r \frac{v^3 - v_{\text{ci}}^3}{v_r^3 - v_{\text{ci}}^3}, & v_{\text{ci}} \leq v < v_r, \\ P_r, & v_r \leq v \leq v_{\text{co}}. \end{cases} \quad (33)$$



**FIGURE 4.** Horizon-wise test metrics for baseline comparison across datasets. Lower is better for RMSE and MAE, and higher is better for  $R^2$ .

For each test sample and horizon  $h \in \{1, \dots, 6\}$ , this study converts  $(v_h, \hat{v}_h)$  into  $(P(v_h), P(\hat{v}_h))$  and computes power-domain RMSE, MAE, and  $R^2$  by aggregating across all test samples and horizons. In addition, this study reports a 6-h window energy error that translates forecast accuracy into energy impact. With  $\Delta t = 1$  h, the true and predicted energies per window are

$$E = \sum_{h=1}^6 P(v_h) \Delta t, \quad \hat{E} = \sum_{h=1}^6 P(\hat{v}_h) \Delta t, \quad (34)$$

and this study reports RMSE, MAE, and  $R^2$  over  $\{(E, \hat{E})\}$  in MWh.

Because (33) clips to 0 outside the operating range and saturates at  $P_r$ , improvements in m/s do not translate uniformly into improvements in MW or MWh. Table 6 therefore reports power and energy proxy metrics for RCHA-MOE and the strongest baseline per dataset selected by energy RMSE.

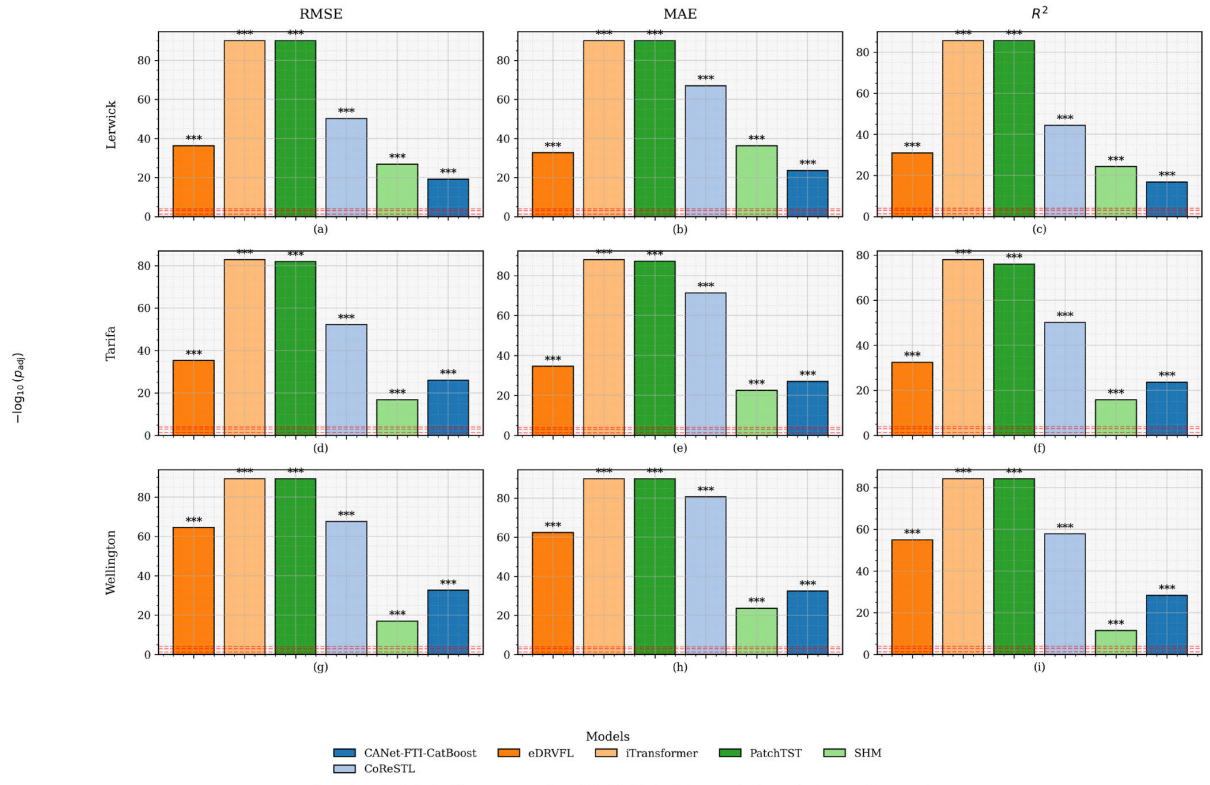
To isolate the high-sensitivity regime, this study evaluates errors restricted to ramp points with true wind speed  $v \in [v_{ci}, v_r]$ . Table 7 reports ramp-region power metrics together with the ramp fraction, which quantifies how

often the test set operates in the region where  $dP/dv$  is largest.

#### IV. LIMITATIONS AND FUTURE WORK

This study targets a controlled setting to isolate the effect of horizon-conditioned, regime-adaptive fusion. The evaluation uses three datasets and one consistent data product, namely NASA POWER reanalysis-derived fields and a fixed  $3 \times 3$  sampling rule [44]. This scope supports clean attribution through a shared tuning budget and strong aggregation baselines, but it limits statistical generalization across climates, terrains, and data sources. In addition, the  $3 \times 3$  neighborhood is constructed at an approximate 100 km offset, which provides mesoscale context but does not resolve microscale terrain speed-up, wake effects, or dataset-specific turbulence that can dominate turbine-level variability. This work therefore does not claim universal optimality for all farms or sensor networks; instead, it positions the reported gains as evidence that inference-time routing is beneficial under a reproducible NASA POWER spatio-temporal benchmark. Future work will extend the evaluation to additional data products, more wind regimes, turbine- or farm-level measurements, different spatial resolutions, and longer forecast horizons to





**FIGURE 5.** Wilcoxon signed-rank tests comparing RCHA-MOE against each baseline using paired batch-level metric estimates. Results are reported as  $-\log_{10}(p_{adj})$  after Benjamini–Hochberg correction, so larger values indicate stronger evidence in favor of RCHA-MOE. Dashed lines mark  $p = 0.05$ ,  $10^{-3}$ , and  $10^{-4}$ .

**TABLE 6.** Power and 6-h energy proxy metrics. Baseline is the strongest method per dataset selected by energy RMSE. Power metrics are in MW and energy metrics are in MWh.

| Dataset    | Model              | Pow RMSE      | Pow MAE       | Pow $R^2$     | Eng RMSE      | Eng MAE       | Eng $R^2$     |
|------------|--------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Lerwick    | RCHA-MOE           | <b>0.5494</b> | <b>0.2527</b> | <b>0.9218</b> | <b>0.9364</b> | <b>0.5984</b> | <b>0.8844</b> |
|            | CANet-FTI-CatBoost | 0.5895        | 0.2768        | 0.9100        | 1.0195        | 0.6500        | 0.8621        |
| Tarifa     | RCHA-MOE           | <b>0.3749</b> | <b>0.1924</b> | <b>0.9359</b> | <b>0.5896</b> | <b>0.3941</b> | <b>0.9109</b> |
|            | SHM                | 0.4045        | 0.2113        | 0.9253        | 0.6525        | 0.4464        | 0.8908        |
| Wellington | RCHA-MOE           | <b>0.5646</b> | <b>0.2881</b> | <b>0.9146</b> | <b>0.9699</b> | <b>0.6577</b> | <b>0.8506</b> |
|            | SHM                | 0.5921        | 0.3070        | 0.9061        | 1.0633        | 0.7210        | 0.8203        |

**TABLE 7.** Ramp-region power metrics (MW) on points with true wind speed  $v \in [v_d, v_r]$ . Ramp fraction reports the percentage of true wind in the ramp. Baseline is the strongest method per dataset selected by ramp RMSE.

| Dataset    | Model              | Ramp frac (%) | Ramp RMSE     | Ramp MAE      | Ramp $R^2$    |
|------------|--------------------|---------------|---------------|---------------|---------------|
| Lerwick    | RCHA-MOE           | 61.3647       | <b>0.6086</b> | <b>0.3622</b> | <b>0.8059</b> |
|            | CANet-FTI-CatBoost | 61.3647       | 0.6525        | 0.3942        | 0.7769        |
| Tarifa     | RCHA-MOE           | 78.1875       | <b>0.3920</b> | <b>0.2276</b> | <b>0.8860</b> |
|            | SHM                | 78.1875       | 0.4164        | 0.2463        | 0.8714        |
| Wellington | RCHA-MOE           | 68.9164       | <b>0.6159</b> | <b>0.3731</b> | <b>0.7970</b> |
|            | SHM                | 68.9164       | 0.6486        | 0.3978        | 0.7748        |

test how the proposed horizon-conditioned routing behaves beyond the present fixed benchmark.

The computational cost of RCHA-MOE is dominated by the three expert forward passes, while the HCRG gate adds only a compact multilayer-perceptron evaluation after the expert predictions are available. Relative to

uniform averaging, the additional inference cost comes mainly from the gate because both methods already evaluate the same three experts. Relative to best-expert selection, RCHA-MOE avoids repeated manual or validation-based reselection during operation because the gate assigns horizon-wise weights directly from the current input descriptor. For the present  $3 \times 3$  patch and  $H = 6$  horizon, this design remains suitable for short-term forecasting because the gate operates on a compact descriptor rather than on the full spatio-temporal tensor. For larger spatial grids, cost will mainly increase through the expert encoders because flattened recurrent and temporal-convolution inputs scale with the number of grid cells, and convolutional recurrent states scale with the spatial map size. For longer horizons, the main increase comes from larger expert output heads and horizon-wise gate weights. Future work will examine sparse spatial neighborhoods, shared expert encoders, model compression, and horizon grouping to preserve scalability under larger grids, longer forecast horizons, and real-time deployment constraints.

## V. CONCLUSION

This study proposes Regime-Conditioned Horizon-aware Mixture-of-Experts (RCHA-MOE) for multi-horizon wind-speed forecasting from a local  $3 \times 3$  spatio-temporal patch. The architecture combines three experts that encode



complementary inductive biases, namely gated recurrence for persistence, temporal dilated convolutions for multi-scale causal structure, and convolutional recurrence for locality-preserving dynamics. The Horizon-Conditioned Regime Gate (HCRG) produces horizon-specific mixture weights conditioned on the input window and fuses expert outputs into a single  $H$ -step forecast.

Experiments on three geographically diverse NASA POWER datasets spanning 2015–2024 at a 1-h sampling rate show that RCHA-MOE achieves the best mean performance across all datasets under a shared tuning and evaluation protocol. Relative to the strongest baseline per dataset, RCHA-MOE improves RMSE by 6.44%–6.93% and MAE by 7.49%–7.65% while increasing  $R^2$  by 0.77%–1.02%. The ablation confirms that uniform averaging is a strong aggregation baseline and that horizon-conditioned gating provides additional gains when expert preferences vary across horizons and regimes, while remaining competitive when simple averaging already matches expert performance. Mapping wind-speed forecasts through a reference power curve further shows that these gains translate into lower power and 6-h energy proxy errors, including in the ramp operating region where power is most sensitive to wind-speed error. Together, these results support regime-conditioned, horizon-aware inference-time fusion as an effective design choice for short-term WSP within this controlled NASA POWER benchmark, while broader validation across additional data sources, spatial resolutions, terrain regimes, and turbine- or farm-level measurements remains an important direction for future work. Future applications include short-term wind-power scheduling, ramp-aware operational monitoring, and lightweight forecasting modules for renewable-energy decision support, provided that the method is further validated under site-level measurements and deployment-specific latency constraints.

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